Hypomorphic homozygous mutations in phosphoglucomutase 3 (*PGM3*) impair immunity and increase serum IgE levels

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Background: Recurrent bacterial and fungal infections, eczema, and increased serum IgE levels characterize patients with the hyper-IgE syndrome (HIES). Known genetic causes for HIES are mutations in signal transducer and activator of transcription 3 *(STAT3)* and dedicator of cytokinesis 8 *(DOCK8)*, which are involved in signal transduction pathways. However, glycosylation defects have not been described in patients with HIES. One crucial enzyme in the glycosylation pathway is phosphoglucomutase 3 (PGM3), which catalyzes a key step in the synthesis of uridine diphosphate N-acetylglucosamine, which is required for the biosynthesis of N-glycans.

Objective: We sought to elucidate the genetic cause in patients with HIES who do not carry mutations in *STAT3* or *DOCK8*.

Methods: After establishing a linkage interval by means of SNPchip genotyping and homozygosity mapping in 2 families with HIES from Tunisia, mutational analysis was performed with selector-based, high-throughput sequencing. Protein expression was analyzed by means of Western blotting, and glycosylation was profiled by using mass spectrometry.

Results: Mutational analysis of candidate genes in an 11.9-Mb linkage region on chromosome 6 shared by 2 multiplex families identified 2 homozygous mutations in *PGM3* that segregated

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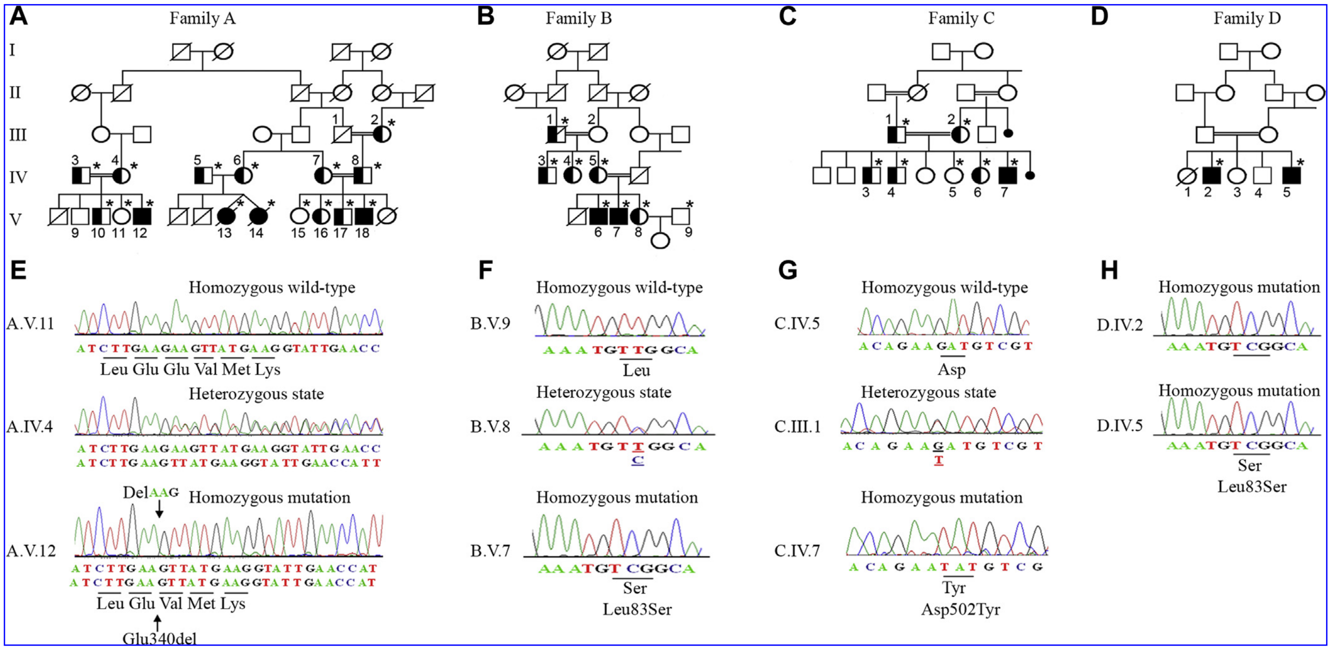

FIG 1. Segregation of *PGM3* mutations with disease status in families with AR-HIES. **A-D,** Family A, p.Glu340del; families B and D, p.Leu83Ser; and family C, p.Asp502Tyr. *Circles*, Female subjects; *squares*, male subjects; *solid symbols*, affected subjects with homozygous mutations; *half-solid symbols*, heterozygous carriers; *open symbols*, healthy members with wild-type *PGM3*; *slashes*, deceased subjects; *double horizontal lines*, consanguinity; *black dots*, miscarriages. *Asterisks* indicate mutations confirmed by using sequencing. **E-H,** Sequence analyses of unaffected subjects without *PGM3* mutations *(top)*, unaffected heterozygous carriers (*middle* in Fig 1, *E-G*), and homozygous affected subjects *(bottom)*.

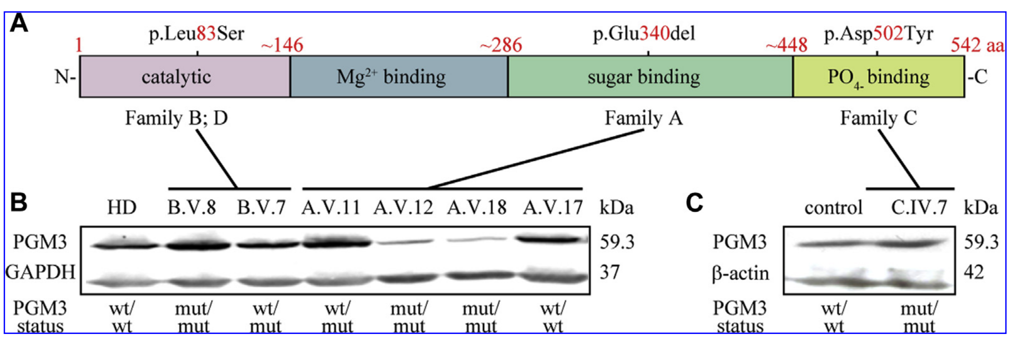

FIG 2. Homozygous p.Glu340del mutations within the sugar-binding domain cause reduced PGM3 expression. **A,** Identified mutations localize to distinct protein domains. *Numbers* indicate amino acid residues. **B,** Western blotting shows reduced PGM3 abundance in EBV-transformed B cells from both homozygous carriers of family A. Mutational status is indicated below each lane. Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) confirms equal loading. *HD*, Healthy donor. **C,** Normal PGM3 expression in immortalized T cells from the patient in family C who had insufficient B-cell numbers. β-actin was used as a loading control.

patients (A.V.12 and C.IV.7), did not affect the glycosylation pattern of transferrin in the blood (see Fig E3 in this article's Online Repository at www.jacionline.org), which is a commonly used diagnostic test for congenital defects of glycosylation.[25] Enzymatic testing revealed that the catalytic activity is reduced in mutant PGM3, depending on the substrate dose, with 20% to 30% of residual activity (Fig 3, *B*). The effect was more pronounced with the p.Glu340del mutation (family A; $P < .0005$ at 12.5 μmol/L GlcNAc-6-P) than with the p.Leu83Ser mutation (family B, $P < .002$). We concluded that the mutant PGM3 retains its catalytic specificity, although with impaired enzymatic activity. In the accompanying article by Zhang et al,[26] a biochemical defect was shown by analyzing intracellular UDP-GlcNAc levels.

PGM3 mutations inhibit the formation of complex N-glycan subtypes

By using a common substrate (UDP-GlcNAc), the GlcNAc transferases I (Michaelis-Menten constant Km 0.04) and II

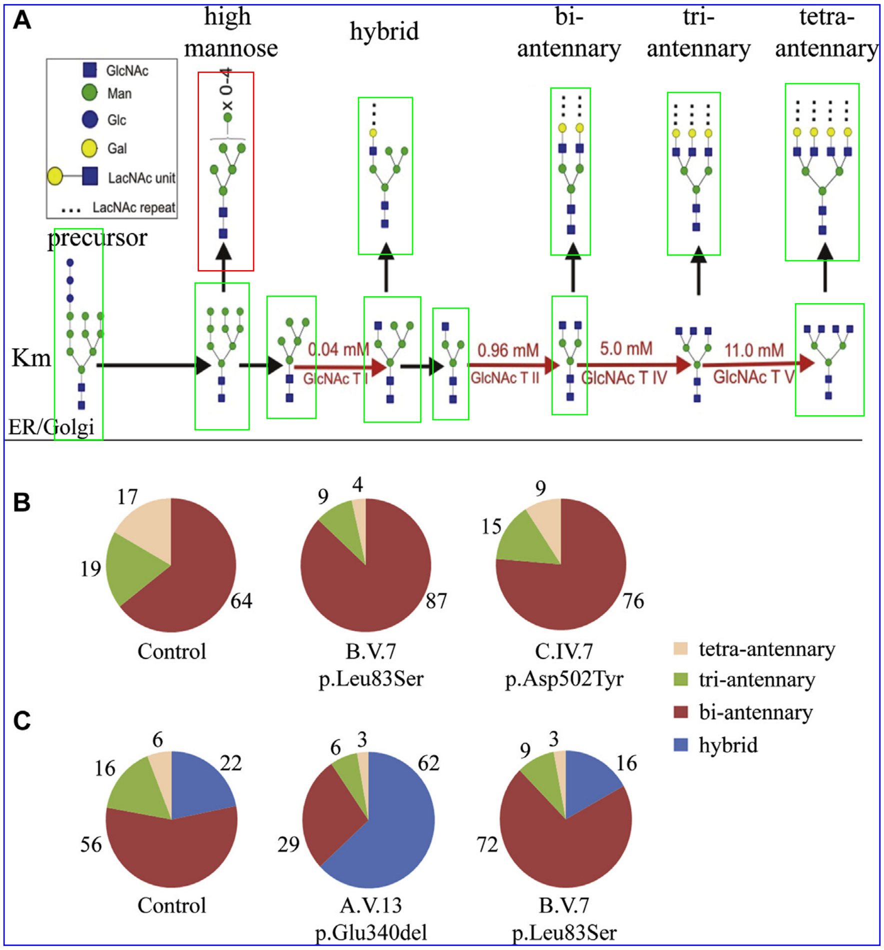

FIG 4. Incomplete assembly of complex-type glycans in patients with *PGM3* mutations. **A,** PGM3 links to N-glycosylation through UDP-GlcNAc *(red)*. The common substrate for assembly of the complex-type N-glycans in the Golgi apparatus *(vertical arrows)* is used by 4 GlcNAc transferases (transferase *[T]* I, II, IV, and V; *red arrows*) with decreasing catalytic activity (increasing Km values [mmol/L UDP-GlcNAc]).[27] UDP-GlcNAc produced by mutant PGM3 might be insufficient to allow for completion of N-glycan antenna formation. *ER*, Endoplasmic reticulum. **B,** Accumulation of bi-antennary N-glycans in neutrophils[28] from patients with *PGM3* mutations. Decreased relative abundance of tri-antennary and tetra-antennary glycans (percentage values of N-glycans detected by using glycomic profiling; see Figs E4 and E5) in affected patients B.V.7 (p.Leu83Ser; catalytic domain) and to a lesser extent in affected patient C.IV.7 (p.Asp502Tyr; phosphate-binding domain). **C,** The homozygous *PGM3* mutation p.Glu340del causes accumulation of the hybrid-type N-glycans, with a simultaneous decrease in bi-antennary, tri-antennary, and tetra-antennary N-glycan levels in EBV B cells from patient A.V.13 (p.Glu340del; sugar-binding domain). B cells from patient B.V.7 (p.Leu83Ser; catalytic domain) have decreased tri-antennary and tetra-antennary N-glycan levels and increased bi-antennary N-glycan levels. Percentage values indicate the relative abundance of glycan types.

Impaired T-cell proliferation, T_H2 skewing, and borderline T_H17 cell numbers in patients with *PGM3* mutations

To test whether the HIES phenotype in our patients with *PGM3* mutations is associated with impaired T-cell function, we analyzed freshly isolated PBMCs using fluorescence-activated cell sorting (FACS) analysis for proliferation and T cell–specific cytokine responses after stimulation *ex vivo* (Tables III and IV and see Figs E1 and E2; see also the accompanying article by Zhang et al[26]). T-cell proliferation to recall antigens (purified protein derivative of tuberculin and tetanus toxoid) was reduced in all samples tested (Table IV). Interestingly, T-cell proliferation after stimulation with PHA or anti-CD3 of family B carrying the missense mutation (p.Leu83Ser) was comparable with that seen in healthy control subjects; however, proliferation in family A, with the more severe mutation (p.Glu340del), was again borderline low.

The serum IgE level increase might be explained by abnormal T_H cell differentiation. Therefore we analyzed T_H1-T_H2 subsets in our patients by using intracellular IL-4 and IFN-γ staining (Table III and see Fig E1). Two of 4 patients tested had grossly increased percentages of T_H2 cells (19% and 27%, respectively), likely contributing to the increased serum IgE levels seen in these patients.

In contrast to the other patients, B cells were absent in patient C.IV.7, and an unusual subpopulation of $CD4^+$ cells was detected, simultaneously expressing T_H1 and T_H2 cytokines, as well as a reduced proportion of forkhead box protein 3–expressing regulatory T cells (data not shown).

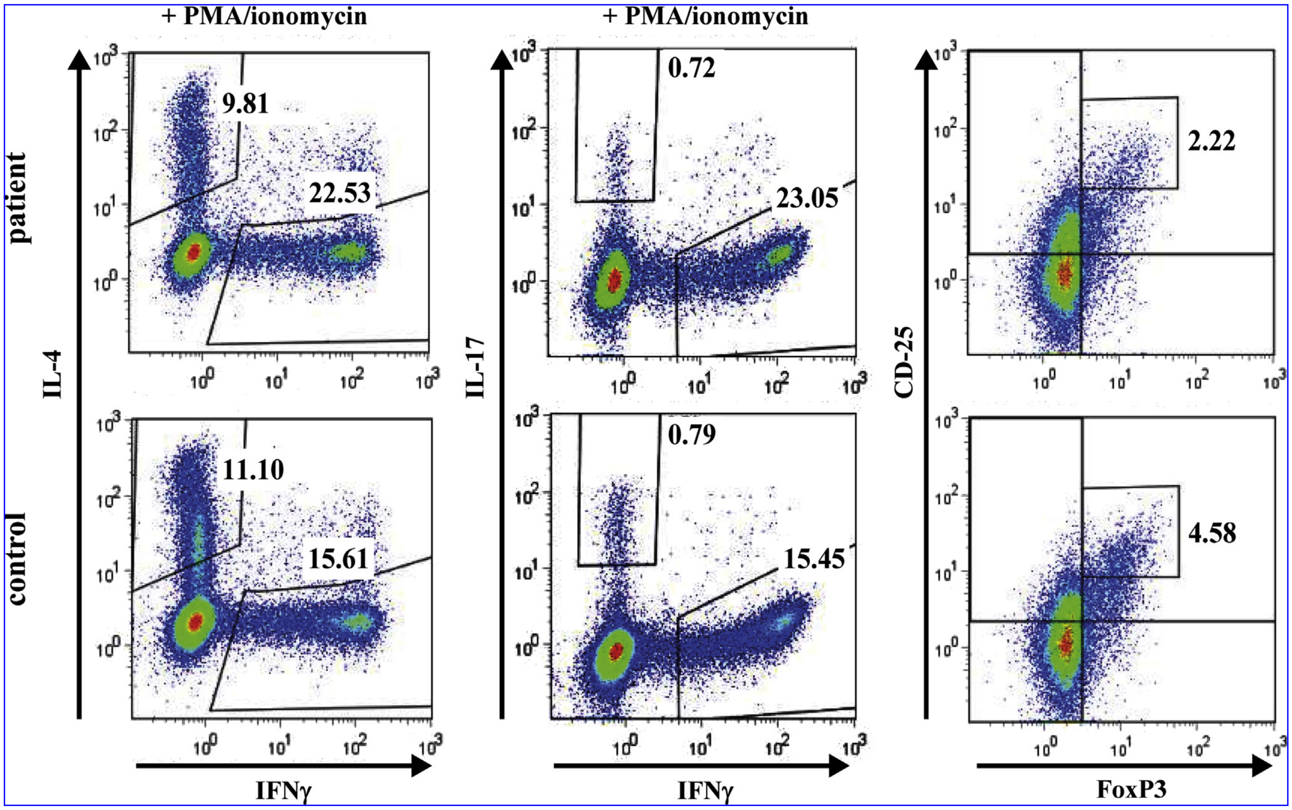

FIG E1. T_H2 skewing but normal T_H17 cell numbers in a patient with *PGM3* mutations. Freshly isolated PBMCs from patient B.V.6 were stimulated, and cytokine profiles were determined by using FACS analysis, as indicated. *Left panels*: *upper left gate*, IL-4–producing T_H2 cells; *lower right gate*, IFN-γ–producing T_H1 cells. *Middle panels*: *upper left gate*, IL-17–producing T_H17 cells; *lower right gate*, IFN-γ–producing T_H1 cells. *Right panels*: Gate shows the proportion of $CD25^{high}$/forkhead box protein 3 *(FoxP3)*$^+$ regulatory T cells.

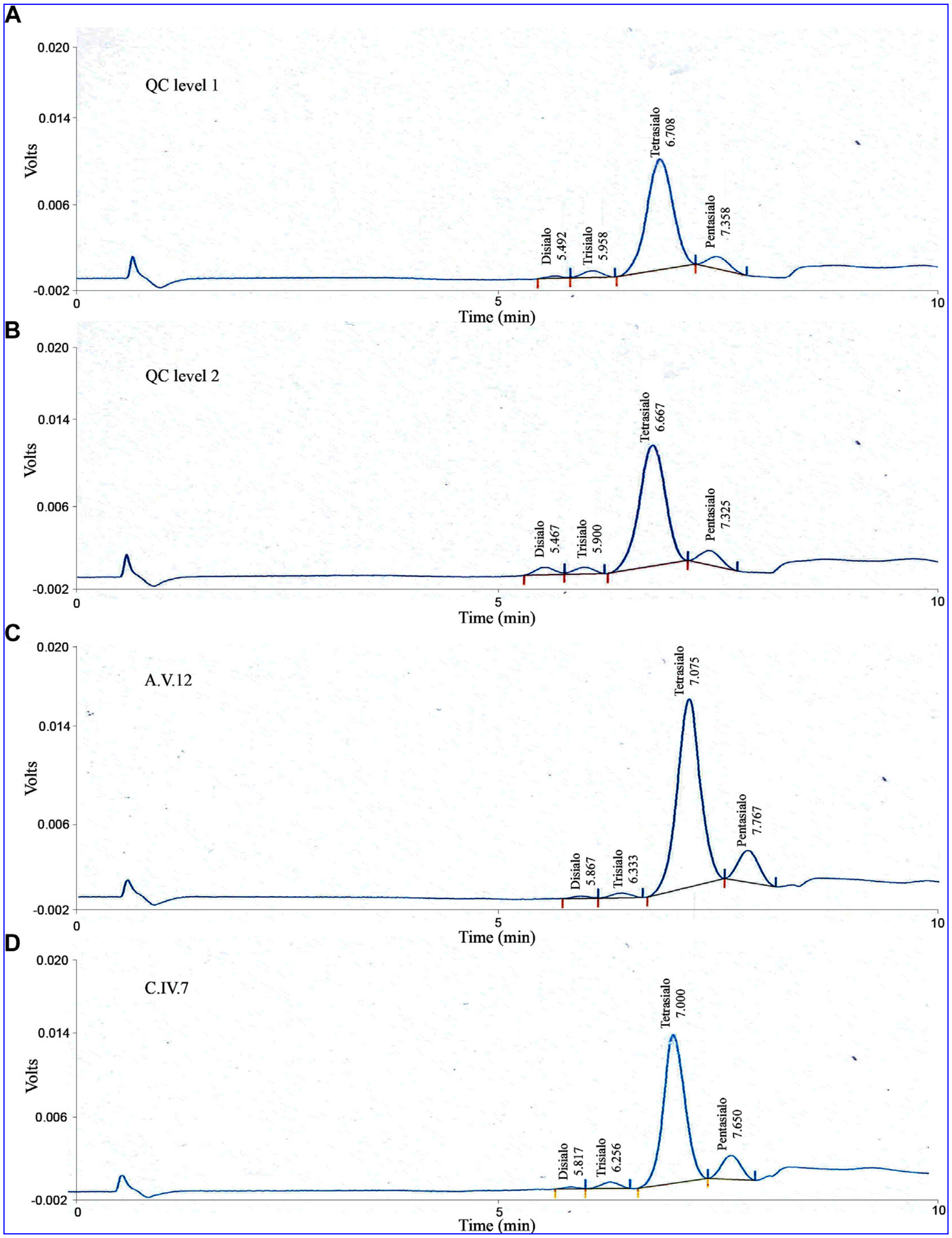

FIG E3. Normal transferrin glycoform patterns in sera from patients with *PGM3* mutations. **A** and **B,** Asialo-, mono-, di-, tri- tetra-, and pentasialo-transferrin glycoform patterns in a control serum were analyzed by using HPLC (quality control *[QC]* level 1 and 2). **C** and **D,** Normal HPLC chromatograms demonstrate normal relative distribution of transferrin glycoforms in sera from patients carrying *PGM3* mutations: A.V.12 (p.Glu340del) and C.IV.7 (p.Asp502Tyr). Transferrin carries 2 bi-antennary N-linked glycans with terminal sialic acid residues, which are used for diagnostic testing for CDGs. Defective transferrin glycosylation would be indicated by an increase of at least one of the asialotransferrin, mono-sialotransferrin, di-sialotransferrin, or tri-sialotransferrin glycoform levels and a decrease in tetra-sialotransferrin levels.

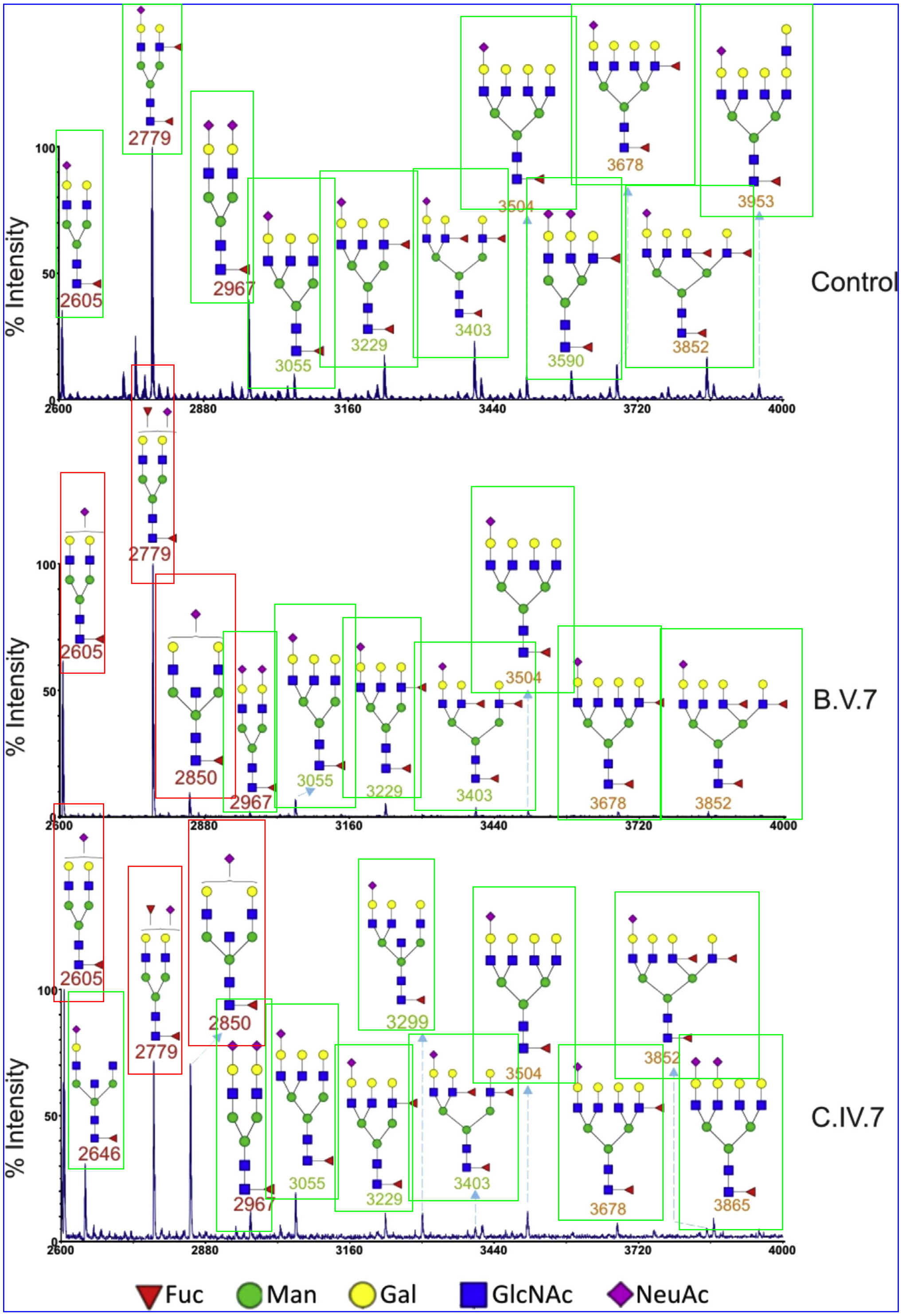

FIG E4. Accumulation of bi-antennary N-glycans in neutrophils from patients with *PGM3* mutations. Mass spectra of permethylated N-glycans of neutrophils from a healthy control subject and the affected subjects B.V.7 and C.IV.7. Results are summarized in Fig 4, *B*, with consistent color codes. Glycomic data of the control is from Babu et al.[E16]